Unit Testing

A "unit" is not merely an isolated function; rather, it consists of a class and its set of methods. Unit testing verifies—in isolation—that an object fulfills its contract, maintains the integrity of its state, and executes its logic correctly.

Unit tests interact strictly with a class's public interface, validating its behavior without needing to violate the data hiding of its private attributes.

Each test instantiates and evaluates an object independently, ensuring that state changes in one test do not affect the others.

It allows for testing how an object interacts with or delegates responsibilities to other objects (collaborators), facilitating the use of simulated objects (mocks) when one class depends on another.

Junit java

JUnit is the standard testing framework in Java that allows for the validation of our classes and objects to be structured using dedicated test classes.

Common JUnit Annotations

JUnit is the standard testing framework in Java that provides a structured, automated mechanism for validating that individual classes, methods, and objects behave according to their intended design specifications through dedicated test classes. It standardizes unit testing by offering a rich set of annotations to manage test execution lifecycles, static assertion methods to verify program correctness against expected outcomes, and seamless integration with modern build tools and IDEs, thereby ensuring code reliability, preventing regression bugs, and supporting maintainable software architecture.

Assertion Methods

These are static methods that allow you to verify whether the actual result returned by the code matches the expected result. If the assertion fails, the test is marked as failed.

Equality assertions evaluate whether two values or primitive results represent equivalent state.

Boolean assertions check whether logical conditions evaluate to true or false.

Nullability assertions verify the presence or absence of an instantiated object reference.